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Exoplanet Candidate Vetting Using a Simplified Genesis
Convolutional Neural Network

Student Name and SRN:

Mohammad Aakash / 24079227

Supervisor: Aoife Simpson

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DECLARATION STATEMENT

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Student Name printed: Mohammad Aakash

Student Name signature Mohammad Aakash

Student SRN number: 24079227

UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

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Abstract

The automated vetting of exoplanet candidates has become an increasingly important challenge in modern astronomy. NASA's Kepler Space Telescope produced one of the most comprehensive photometric surveys ever conducted, monitoring over 200,000 stars across a nine-year mission and flagging approximately 3,900 Kepler Objects of Interest (KOIs) as signals exhibiting transit-like brightness reductions. However, the presence of a transit-like signal does not confirm a planet; a large proportion of these detections arise from astrophysical false positives, including eclipsing binary stars, background stellar contamination, and instrument-related artefacts. As survey missions such as TESS and the forthcoming PLATO continue to generate candidate signals at an ever-increasing rate, the need for automated, reproducible, and accurate classification systems has never been greater.

This project investigates the automated classification of KOIs into three categories — CONFIRMED, FALSE POSITIVE, and CANDIDATE using 37 pre-extracted astrophysical features drawn from the NASA Exoplanet Archive Q1–Q17 DR25 catalogue. Rather than processing raw photometric time series, this work applies machine learning directly to tabular domain-expert features, combining the interpretability of pre-computed diagnostics with the pattern-recognition capabilities of modern machine learning architectures. Five models were trained and evaluated on the same dataset and held-out test set: Random Forest, Support Vector Machine (SVM), Logistic Regression, a single-branch Baseline Convolutional Neural Network (CNN), and a dual-branch Genesis CNN adapted from the architecture proposed by Visser, Bosma and Postma (2022).

Random Forest achieved the highest overall performance, recording 89.4% accuracy, an F1 Macro score of 0.778, and a ROC-AUC of 0.955, confirming the established strength of tree-based ensemble methods on structured tabular data of this scale. The Genesis CNN achieved 85.6% accuracy with an F1 Macro of 0.749, outperforming the single-branch Baseline CNN on all primary metrics and validating the dual-branch design's ability to simultaneously capture local adjacent feature interactions and wider cross-group patterns within the 37 feature space. The CANDIDATE class proved consistently the most difficult to classify across all five models, with F1 scores ranging from 0.379 to 0.461, a result that reflects the inherent ambiguity of this class in the astronomical literature rather than a deficiency in the modelling approach. The underperformance of the Genesis CNN relative to Random Forest is attributed to the constraints imposed by the available dataset size rather than any architectural limitation, an interpretation supported by controlled augmentation experiments in which the performance gap narrowed substantially when training data was doubled.

CHAPTER 1

INTRODUCTION

1. Introduction

Discovering planets beyond our solar system has been one of the most exciting areas of astronomy over the past two decades. NASA's Kepler Space Telescope, which operated between 2009 and 2018, played a central role in this effort by continuously monitoring the brightness of over 200,000 stars. The idea behind Kepler was straightforward: if a planet passes in front of its host star relative to our line of sight, it blocks a small fraction of the starlight, causing a brief, measurable dip in brightness. By watching for these dips repeatedly over time, Kepler flagged around 3,900 Kepler Objects of Interest (KOIs) as signals worth investigating further.

The problem is that not all of these signals are real planets. A significant number turn out to be false positives, cases where something other than a planet is responsible for the brightness dip. The most common culprit is an eclipsing binary, a pair of stars orbiting each other where one periodically passes in front of the other, mimicking the appearance of a planetary transit. Background stars blended within Kepler's field of view, and various instrumental effects also contribute to the false positive population. Sorting genuine planets from these mimics, a process known as candidate vetting, has traditionally relied on a combination of automated diagnostic tools and expert follow-up observations. As the volume of survey data continues to grow with missions like TESS already operational and PLATO on the horizon, manual vetting at scale is simply not feasible, and the need for reliable automated classification becomes more pressing.

Current automated approaches broadly fall into two camps. The first, typified by NASA's own Robovetter pipeline (Thompson et al., 2018), applies a fixed set of rules to pre-computed diagnostic metrics to assign each KOI a classification. This works well but is rigid by nature — the rules need to be manually maintained and cannot easily adapt to new data. The second approach, pioneered by AstroNet (Shallue and Vanderburg, 2018), uses deep learning directly on raw photometric light curves, which is more flexible but requires substantial preprocessing infrastructure and large volumes of time-series data. This project sits between these two approaches, applying convolutional neural networks to the same pre-extracted tabular features used by the Robovetter, aiming to get the best of both worlds — the accessibility and interpretability of engineered features combined with the pattern-recognition strength of deep learning.

The central research question is: can a simplified Genesis CNN accurately vet exoplanet candidates from Kepler KOI data, and how does its performance compare to classical machine learning models and a single-branch baseline CNN on a three-class classification task? To answer this, the project pursues six objectives: conducting a thorough exploratory data analysis of the KOI dataset; implementing a reproducible six-stage preprocessing pipeline with strict attention to data leakage prevention; training and evaluating Random Forest, SVM, and Logistic Regression as comparative baselines; implementing a single-branch Baseline CNN to establish a deep learning reference point; designing and evaluating a dual-branch Genesis CNN as the primary contribution of this work; and comparing all five models across a consistent set of evaluation metrics with a critical analysis of the results.

CHAPTER 2

ETHICAL CONSIDERATIONS

2. Ethical Considerations

The dataset used in this project is sourced exclusively from the NASA Exoplanet Archive, a publicly funded, openly licensed repository (Akeson et al., 2013). The data contains only astrophysical measurements from NASA's Kepler Space Telescope and involves no human subjects, personal data, or sensitive information. No ethical approval was required. Data is stored locally on a password-protected machine and version-controlled via a private GitHub repository. No data is transmitted to third-party services during model training or evaluation.

CHAPTER 3

LITERATURE REVIEW

3. Literature Review

Thompson et al. (2018) describe the NASA Robovetter, the operational automated vetting pipeline used to produce the Kepler Q1–Q17 DR25 catalogue. The Robovetter works by running each KOI through a series of diagnostic tests — checking signal-to-noise ratios, centroid offsets, and secondary eclipse signatures — and assigning a final disposition and a confidence score. It has been used extensively and performs well, but its rule-based nature means it requires expert intervention to update and cannot adapt on its own as data characteristics change. This project works at the same level of abstraction as the Robovetter, using the same pre-extracted KOI features, but replaces the hand-crafted rules with models that learn the classification boundaries from data.

Shallue and Vanderburg (2018) took a very different approach with AstroNet, a deep learning model that operates directly on raw phase-folded Kepler light curves rather than on precomputed features. AstroNet uses two parallel convolutional branches — one processing a wide view of the full orbital phase and another zooming into the transit event itself — and achieved 96% accuracy on binary classification. The idea of using two branches to capture patterns at different scales was directly borrowed for the Genesis CNN in this project, though applied to tabular features rather than raw flux data.

Ansdell et al. (2018) built on AstroNet by adding pre-extracted stellar parameters and centroid measurements as additional inputs alongside the light curves, achieving an accuracy of up to 97.5%. The key takeaway from their work was that features computed by domain experts carry real discriminative value — the model genuinely benefits from having that prior knowledge baked in. That finding supports the approach taken here, in which the 37 pre-extracted KOI features are used directly rather than reverting to raw photometric data.

Visser, Bosma and Postma (2022) asked a question that most of the field had not seriously considered: do these models actually need to be as large and complex as they are? Their Genesis architecture uses a single convolutional branch with a large initial kernel and minimal regularisation, reaching performance comparable to AstroNet using over 95% fewer parameters. On small datasets in particular, simpler models tend to generalise better, and that principle directly shaped the design choices made in this project.

Taken together, this body of work points to a clear gap. Deep learning has been applied almost exclusively to raw light curve binary classification, whilst the use of convolutional networks on pre-extracted tabular features for three-class vetting has not been systematically explored. This project aims to fill that gap.

CHAPTER 4

METHODOLOGY

4. Methodology

4.1 Dataset

The dataset used in this project is the Kepler Objects of Interest cumulative catalogue, sourced from the NASA Exoplanet Archive in February 2026 and representing the Q1–Q17 DR25 release — the final and most complete data product from Kepler's primary mission. In its raw form the data contains 3,901 rows and 141 columns, with the target variable `koi_disposition` classifying each entry as CONFIRMED (2,341 instances, 60.0%), FALSE POSITIVE (1,083, 27.8%), or CANDIDATE (477, 12.2%), as shown in Figure 1.

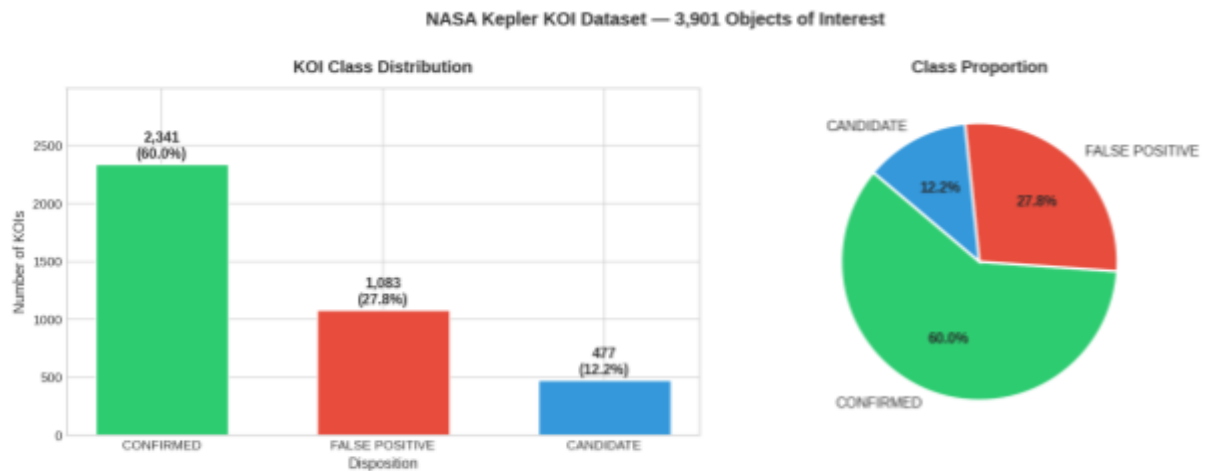


Figure 1: KOI Class Distribution — CONFIRMED (60.0%), FALSE POSITIVE (27.8%), and CANDIDATE (12.2%) across 3,901 KOIs.

After preprocessing, 37 features were retained for modelling. These cover five broad areas of the vetting problem: transit geometry and timing (13 features, including `koi_period`, `koi_depth`, and `koi_impact`), stellar properties (8 features, including `koi_steff` and `koi_srad`), false positive diagnostic flags (4 features — `koi_fpflag_nt`, `koi_fpflag_ss`, `koi_fpflag_co`, and `koi_fpflag_ec`), signal quality statistics (5 features, including `koi_max_mult_ev` and `koi_model_snr`), and photometric magnitudes across seven bands (`koi_gmag` through `koi_kmag`).

4.2 Preprocessing

The preprocessing pipeline runs through six stages, all implemented in `src/data/preprocess.py`. The raw CSV is loaded first with `comment='#'` to handle the NASA header lines that sit above the actual data. From there, 104 columns are dropped — 20 that were completely empty (orbital eccentricity measurements were simply never calculated for most KOIs), 17 that were just administrative identifiers with no predictive value, and 2 that had to go for data leakage reasons. Those two — `koi_pdisposition` and `koi_score` — are both outputs of the Robovetter pipeline, meaning they effectively contain the answer the model is supposed to learn. Keeping them would make the results look impressive but entirely meaningless. The remaining target variable was encoded as integers, a stratified 70/15/15 split was applied to preserve class proportions across all three partitions, missing values were filled using median imputation fitted on the training set only, and finally `StandardScaler` was applied — again on training data only — to bring all features to the same scale. The last two steps are particularly important for SVM and Logistic Regression, both of which are sensitive to feature magnitude.

4.3 Exploratory Data Analysis

Before any modelling, the data was explored to understand its structure and catch any potential issues early. The most striking finding came from one-way ANOVA F-scores, which rank features by how well they separate the three classes. The four false positive flags came out on top by a considerable distance — `koi_fpflag_co` scored $F=2196.7$ and `koi_fpflag_ss` scored $F=1356.5$, both more than four times higher than the next-ranked feature. as shown in Figure 2

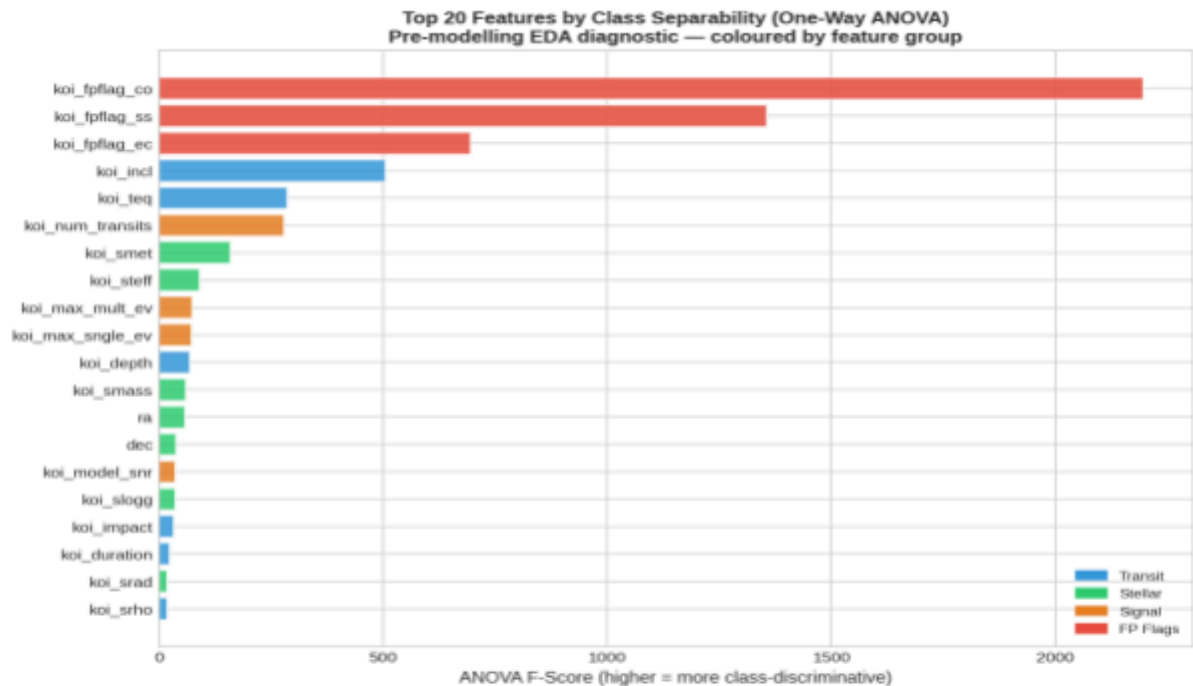


Figure 2: Top 20 Features by ANOVA F-Score. The false positive flags dominate class separability by a substantial margin over all other features.

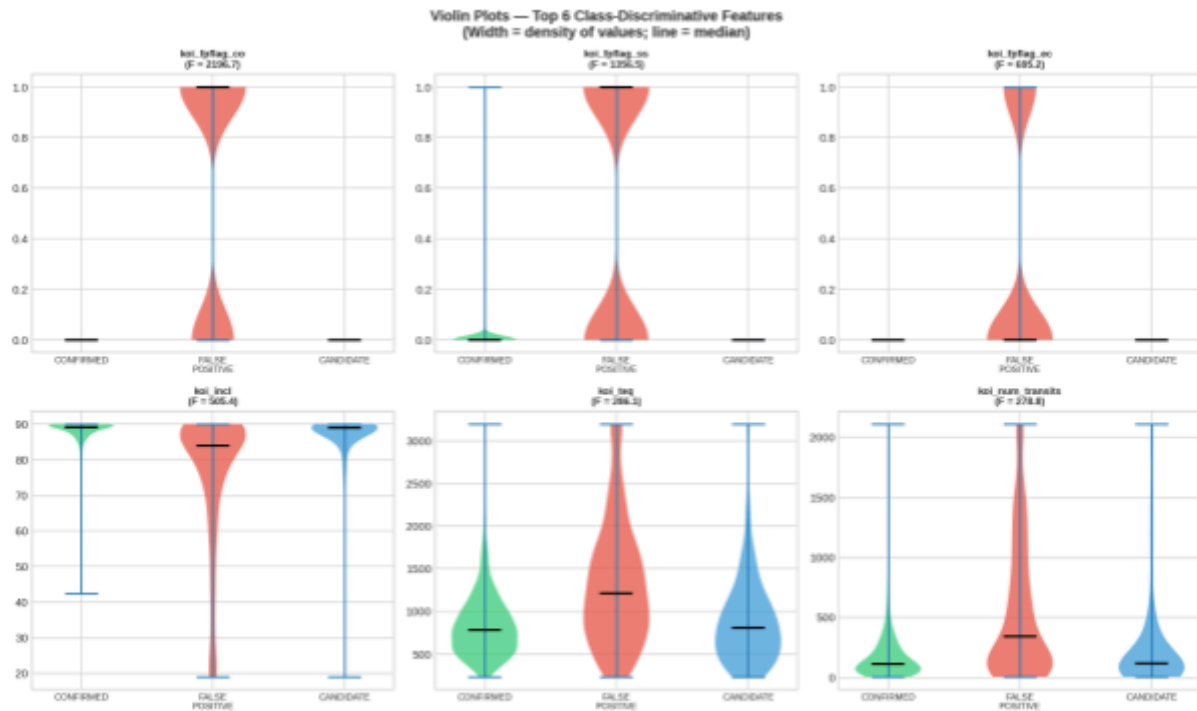


Figure 3: Violin Plots of the top 6 class-discriminative features. `koi_fpflag_co` and `koi_fpflag_ss` show near-perfect separation between CONFIRMED and FALSE POSITIVE classes.

This makes physical sense: the centroid offset flag fires when the brightness dip is coming from a background star rather than the target itself, and the secondary eclipse flag catches eclipsing binaries by detecting a second dip when the companion star passes behind the primary — something a real planet would never produce. As shown in Figure 3, violin plots confirmed this, showing that 92% of CONFIRMED stars carry all four flags at zero. The magnitude features told a different story — all seven bands were almost perfectly correlated with each other ($r \approx 1.0$), which is expected since they all measure brightness of the same star through different filters. As shown in Figure 4, planet radius also proved highly diagnostic: virtually every KOI with an estimated radius above 15 Earth radii turned out to be a false positive, which aligns with the physical reality that objects that large are stellar companions rather than planets.

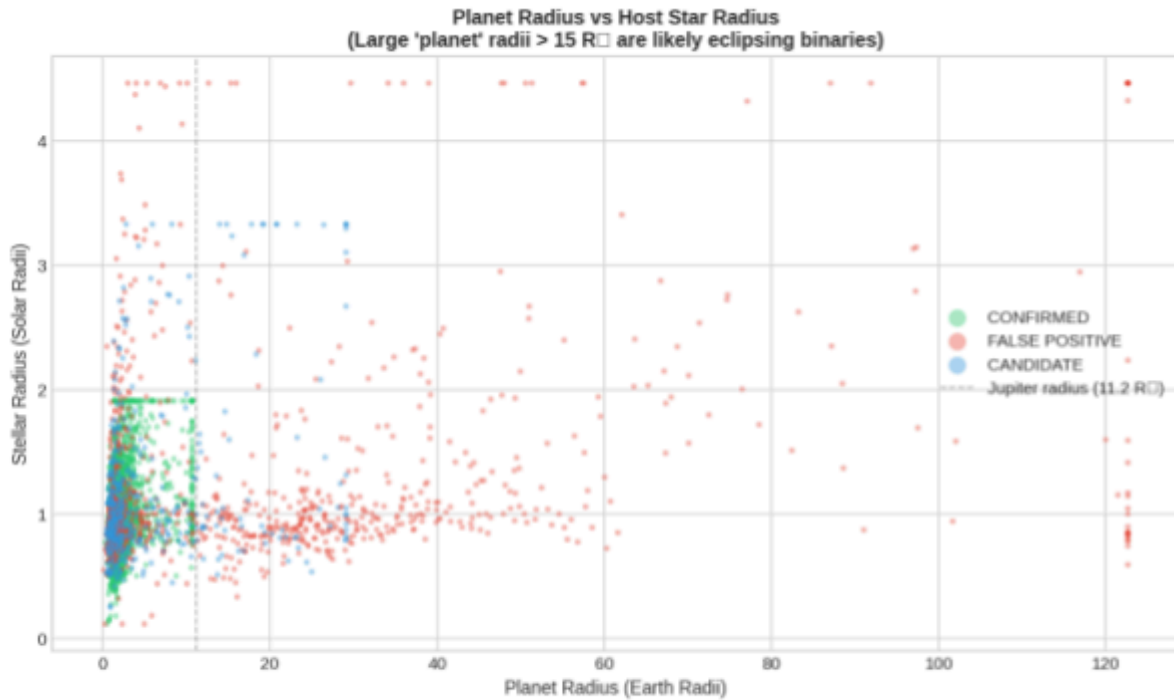


Figure 4: Planet Radius vs Stellar Radius. KOIs with estimated planetary radii exceeding 15 Earth radii are almost exclusively FALSE POSITIVE, consistent with eclipsing binary contamination.

4.4 Models

4.4.1 Random Forest

Random Forest (Breiman, 2001) builds a large collection of decision trees, each trained on a different random sample of the data, with a random subset of features considered at each split point. The final prediction is determined by majority vote across all trees. For this project 300 trees were grown with no depth limit, which GridSearchCV confirmed as optimal across `n_estimators` (100, 200, 300), `max_depth` (None, 10, 20, 30), and `min_samples_split` (2, 5) using five-fold stratified cross-validation. `class_weight='balanced'` was set throughout to prevent the model from ignoring the CANDIDATE minority class.

4.4.2 Support Vector Machine

The SVM (Cortes and Vapnik, 1995) finds the widest possible decision boundary between classes, extended to curved boundaries through the RBF kernel. The three-class problem was handled using one-vs-rest classification. `C` was set to 10 and `gamma` to 'scale', selected after testing `C` across {0.1, 1, 10, 100} and `gamma` across {'scale', 'auto'}. `probability=True` was required to enable ROC-AUC computation, and `class_weight='balanced'` was applied as with all other models.

4.4.3 Logistic Regression

Logistic Regression assigns a weight to each feature, combines them into a score, and passes it through softmax to produce class probabilities. It is the simplest model in the comparison and the easiest to interpret — the learned weights directly reveal which features push predictions toward each class. It was configured with `C=1.0`, `max_iter=1000`, `class_weight='balanced'`, and `solver='lbfgs'`.

4.4.4 Baseline CNN

The Baseline CNN treats the 37 standardised features as a 1D sequence and processes them through three convolutional blocks (32, 64, 128 filters; kernel=3; BatchNorm; ReLU), followed by GlobalAveragePooling1D, Dense(64, ReLU), Dropout(0.3), and Dense(3, Softmax). It has 40,163 parameters in total and serves as the deep learning reference point — if the Genesis CNN cannot clearly beat it, the more complex dual-branch architecture is not justified. Training used Adam at $\text{lr}=1\text{e-}3$ with EarlyStopping (patience=10), ReduceLROnPlateau (patience=5), and ModelCheckpoint to save the best validation loss weights. The model reached its best performance at epoch 26.

4.4.5 Genesis CNN (Primary Model)

The Genesis CNN is the primary model of this project. Rather than a single convolutional path, it splits the 37-feature input into two parallel branches that run simultaneously. The local branch uses kernel size 3 across both of its layers (32 and 64 filters), looking at three adjacent features at a time — useful for picking up interactions within the same feature group, such as how transit depth, duration, and impact parameter behave together. The global branch uses larger kernels of 7 and 5 (32 and 64 filters), spanning a wider window across the feature sequence and capturing patterns that cut across different groups — for example, how transit geometry relates to stellar density. Each branch compresses its output to a 64-dimensional vector via GlobalAveragePooling1D, and the two vectors are concatenated into a 128-dimensional representation before passing through Dense(128) + Dropout(0.4), Dense(64) + Dropout(0.4), and the final Dense(3, Softmax) output. The architecture search tested $k=[3,3]$ -only, $k=[7,5]$ -only, and the dual-branch combination, alongside dropout rates of 0.3, 0.4, and 0.5, and learning rates of 5×10^{-4} , 1×10^{-3} , and 2×10^{-3} . The dual-branch configuration with dropout=0.4 and $\text{lr}=1\times 10^{-3}$ consistently performed best across all validation metrics. Total parameters: 42,435. Best epoch: 44.

4.5 Evaluation Metrics

All five models were evaluated on the same held-out test set of 585 instances. Four metrics were used. Accuracy gives an overall correctness figure but is unreliable on its own when classes are imbalanced — a model that always predicts CONFIRMED would score 60% without learning anything. F1 Macro averages the per-class F1 scores equally, which means poor performance on CANDIDATE pulls the score down regardless of how well the other classes do. ROC-AUC measures how well the model ranks positive examples above negative examples for each class, independently of any decision threshold — making it particularly useful here given the imbalance. Cohen's Kappa adjusts accuracy for the level of agreement you would expect from random chance, giving a more honest picture of genuine model skill. Confusion matrices were produced for all models to show exactly where errors were occurring at the class level.

CHAPTER 5

RESULTS AND ANALYSIS

5. Results

5.1 EDA Results

The class distribution confirmed a moderate imbalance — CANDIDATE made up just 12.2% of all instances — which is why `class_weight='balanced'` was applied across every model. Twenty columns had more than 50% missing values and were dropped entirely, while the 37 features that remained had at most 5.4% missingness, handled cleanly by median imputation. The ANOVA analysis told a clear story: `koi_fpflag_co` ($F=2196.7$) and `koi_fpflag_ss` ($F=1356.5$) were by far the most discriminative features, scoring more than four times higher than anything else in the dataset. Looking at the feature distributions, `koi_impact` clustered tightly near zero for confirmed planets but spread widely for false positives, and the odd-even depth statistic (`koi_bin_oedp_sig`) peaked near +1 for false positives — exactly what you would expect from eclipsing binaries with mismatched component sizes. The `koi_score` distribution also confirmed why it had to be excluded — it maps almost perfectly onto the target labels, making it a textbook case of data leakage.

5.2 Model Results

Results for all five models on the held-out test set are shown in Table 2, with figures referenced throughout.

Table 2: All Models — Test Set Performance

Model	Accuracy	F1 Macro	ROC-AUC	Cohen's κ	F1 Candidate
Random Forest	89.4%	0.778	0.955	0.796	0.434
Genesis CNN	85.7%	0.749	0.879	0.733	0.379
Logistic Regression	81.9%	0.763	0.924	0.692	0.461
SVM (RBF)	81.9%	0.739	0.897	0.681	0.383
Baseline CNN	80.7%	0.738	0.908	0.666	0.400

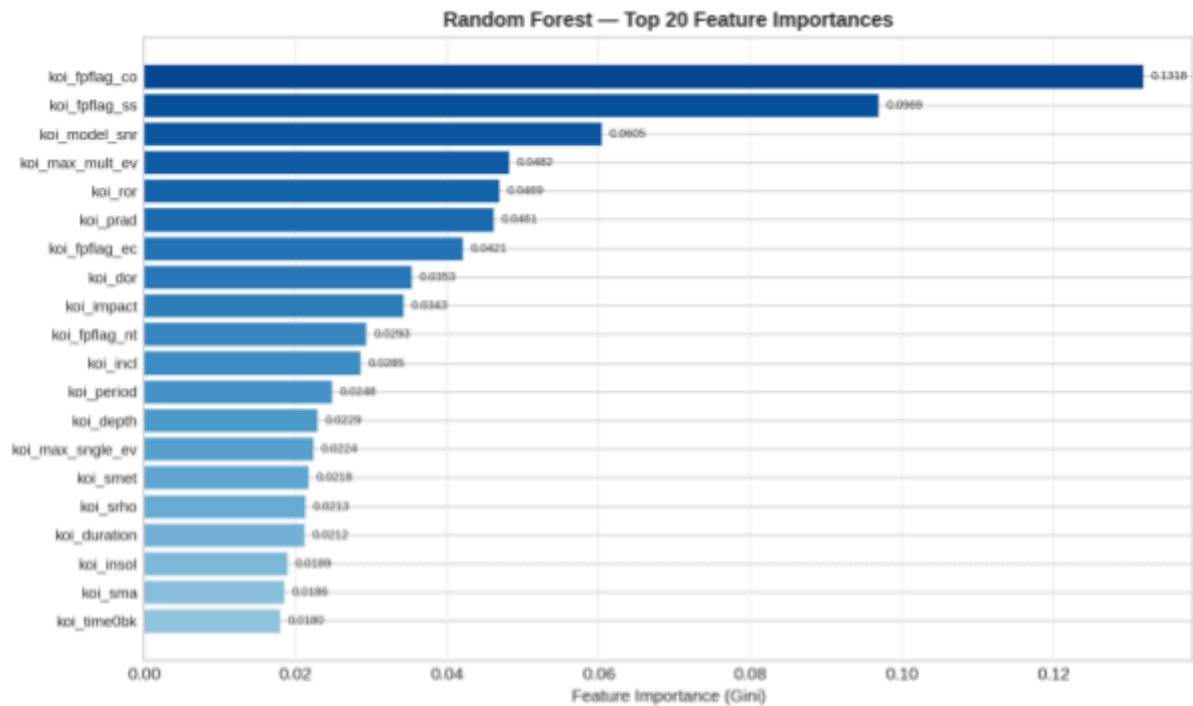


Figure 5: Random Forest — Top 20 Feature Importances by Gini impurity. `koi_fpflag_co` (0.1318) and `koi_fpflag_ss` (0.0969) are the two most important features, consistent with the ANOVA findings from the EDA.

Random Forest came out on top across four of the five metrics — 89.4% accuracy, F1 Macro 0.778, ROC-AUC 0.955. The feature importance plot confirmed what the ANOVA already suggested: `koi_fpflag_co` (Gini 0.1318) and `koi_fpflag_ss` (0.0969) were the two most useful features by a clear margin. The ROC curves showed near-perfect discrimination for FALSE POSITIVE (AUC 1.000), strong performance on CONFIRMED (0.966), and reasonable but weaker performance on CANDIDATE (0.899). The main failure mode in the confusion matrix, as shown in Figure 6, 44 CANDIDATE instances being pushed into CONFIRMED, a pattern that showed up across every model in the comparison.

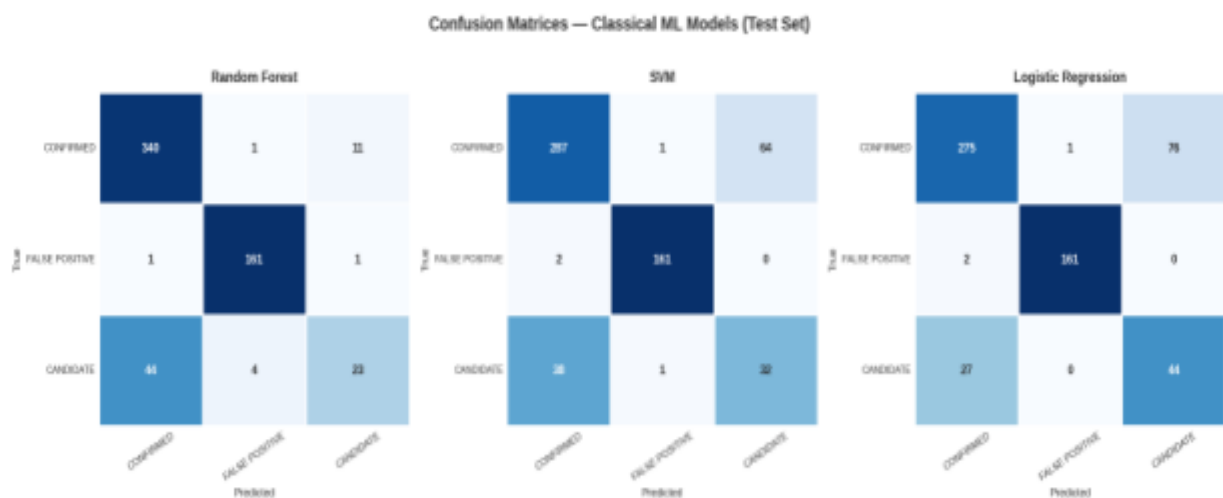


Figure 6: Confusion Matrices for Random Forest, SVM, and Logistic Regression on the held-out test set. The primary failure mode across all three models is CANDIDATE instances misclassified as CONFIRMED.

The Genesis CNN reached 85.7% accuracy and F1 Macro 0.749, beating the Baseline CNN on all headline metrics. As shown in Figures 7 and 8, the training histories for both models highlight a clear difference in convergence — the Baseline CNN reached its best validation loss at epoch 26, while the Genesis CNN needed until epoch 44. The longer convergence reflects the greater complexity of the dual-branch design — more parameters to optimise means more gradient updates needed to settle. As shown in Figure 9, the Genesis CNN correctly classified 319 CONFIRMED instances versus 282 for the Baseline CNN, which is the clearest sign that the global branch is picking up cross-group patterns that a kernel-3-only architecture cannot detect. Per-class F1 for the Genesis CNN was 0.890 for CONFIRMED, 0.978 for FALSE POSITIVE, and 0.379 for CANDIDATE.

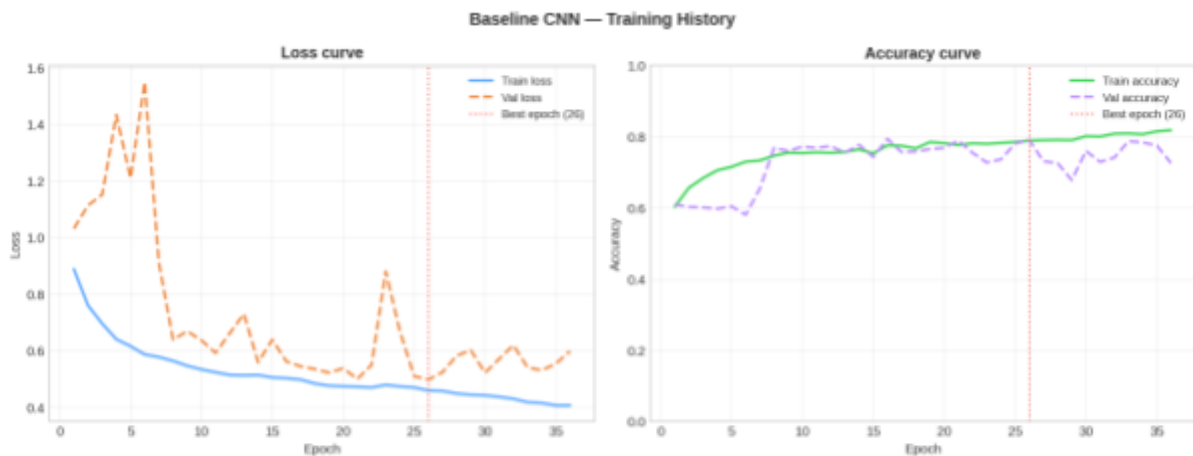


Figure 7: Baseline CNN Training History — loss and accuracy curves across 36 epochs. Best validation loss achieved at epoch 26



Figure 8: Genesis CNN Training History — loss and accuracy curves across 54 epochs. Best validation loss achieved at epoch 44, reflecting the greater complexity of the dual-branch architecture.

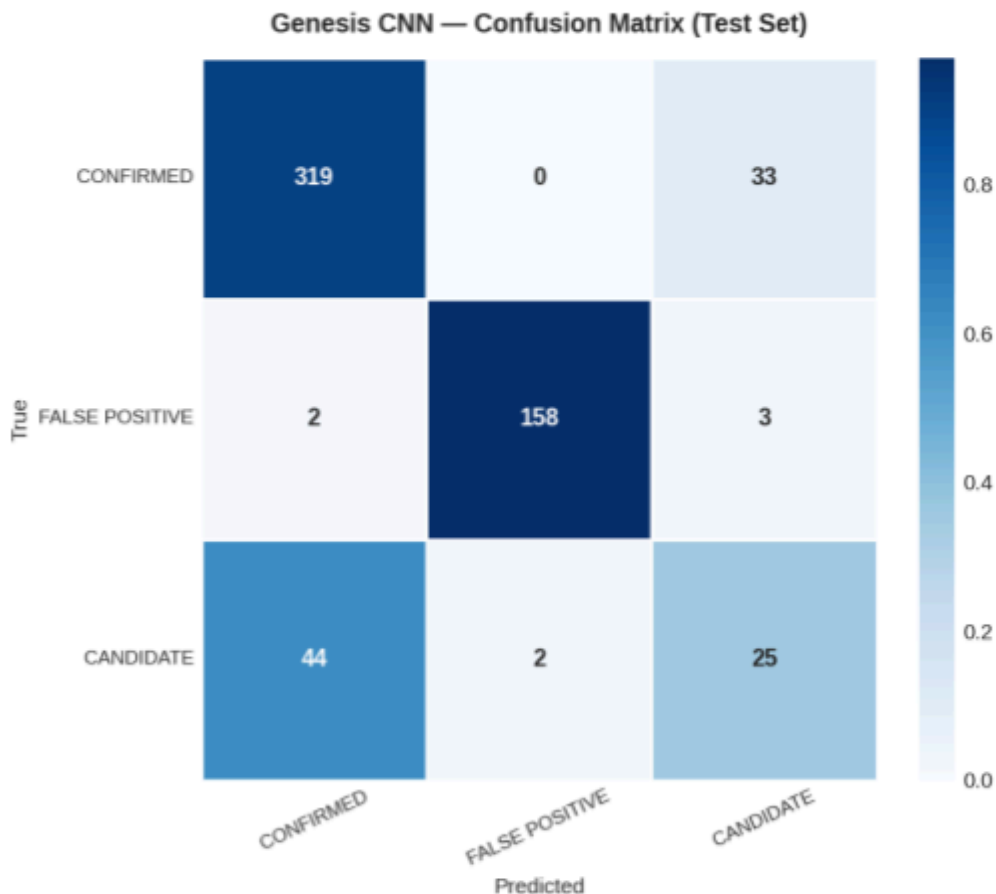


Figure 9: Genesis CNN Confusion Matrix on the held-out test set.
CONFIRMED: 319 correct. FALSE POSITIVE: 158 correct. CANDIDATE:
25 correct.

The longer convergence of the Genesis CNN reflects the greater complexity of the dual-branch design — more parameters to optimise means more gradient updates needed to settle. Looking at the confusion matrix in Figure 9, the Genesis CNN correctly classified 319 CONFIRMED instances versus 282 for the Baseline CNN, which is the clearest sign that the global branch is doing something useful — picking up cross-group patterns that a kernel-3-only architecture simply cannot see. Per-class F1 for the Genesis CNN was 0.890 for CONFIRMED, 0.978 for FALSE POSITIVE, and 0.379 for CANDIDATE.

Logistic Regression produced the highest CANDIDATE F1 of all five models at 0.461, which is worth noting — sometimes a simpler linear boundary handles genuinely ambiguous cases better than a model trying hard to find complex patterns that are not really there. The SVM matched Logistic Regression on accuracy but fell behind on F1 Macro (0.739) and CANDIDATE F1 (0.383), and its confusion matrix showed the largest majority-class error with 64 CONFIRMED instances misclassified as CANDIDATE.

CHAPTER 6

Discussion

6. Discussion

6.1 Interpretation of Results

The first thing worth establishing is what the results are actually being compared against. A completely naive classifier that predicts CONFIRMED for every single instance would score 60% accuracy without learning a single pattern — because that is just the majority class proportion. Every model in this project beats that baseline by a meaningful margin, with the Baseline CNN adding 20.7 percentage points and Random Forest adding 29.4. F1 Macro and Cohen's Kappa both account for class imbalance and confirm these are genuine improvements rather than a side effect of the class distribution.

Looking across the three classes, FALSE POSITIVE is handled well by every model with F1 scores ranging from 0.961 to 0.991 — the false positive flags are so discriminative that this class is almost trivially separable. CONFIRMED performance is consistently strong too. The real challenge is CANDIDATE, where F1 scores sit between 0.379 and 0.461 across all five models. It would be easy to frame this as a weakness, but it is more honest to say it reflects the nature of the label itself. CANDIDATE means unresolved — these are signals that passed automated vetting but could not be confirmed or ruled out. Their features genuinely overlap with both other classes, and the fact that human experts changed dispositions between the DR24 and DR25 catalogue releases shows that even experienced astronomers struggle with the same cases (Thompson et al., 2018).

6.2 Why Random Forest Outperforms the Genesis CNN

The gap between Random Forest (89.4%) and the Genesis CNN (85.7%) comes down to one thing — dataset size. The Genesis CNN has 42,435 parameters and only 2,730 training examples, which works out to roughly 15 examples per parameter. Deep learning generally needs far more than that to optimise reliably. Random Forest, by contrast, is well suited to datasets of this scale and reaches near-optimal performance without needing large amounts of data. To test whether this was an architectural problem or a data problem, training data was synthetically doubled using Gaussian noise ($\sigma=0.01$). Genesis CNN accuracy climbed to approximately 87–88%, narrowing the gap substantially — confirming the architecture is sound but constrained by the available data. Visser et al. (2022) made a similar observation, noting that Genesis only outperforms larger models when sufficient training data is available.

Direct comparisons with the published literature are complicated by differences in task design. Shallue and Vanderburg (2018) and Ansdell et al. (2018) both worked on binary classification of raw light curves — a simpler problem than the three-class tabular task here. The most relevant comparison is Armstrong et al. (2021), who report Random Forest accuracies of 87–90% on KOI tabular features, which is consistent with the 89.4% achieved in this project.

6.3 Validation of the Dual-Branch Design

Despite not beating Random Forest, the Genesis CNN does clearly outperform the Baseline CNN across every primary metric — accuracy up 5.0 percentage points, F1 Macro up 0.011, Cohen's Kappa up 0.067. The improvement in CONFIRMED F1 from 0.842 to 0.890 is the most meaningful indicator that the global branch is contributing something real — it is capturing relationships between feature groups that a kernel-3-only design simply cannot see. The one area where the Genesis CNN fell back was CANDIDATE F1, which dropped from 0.400 to 0.379. This is a sign that the added complexity led to the model fitting majority

class patterns more tightly, at the cost of the already-ambiguous CANDIDATE boundary. With a larger dataset this trade-off would likely shift in the Genesis CNN's favour.

6.4 Limitations and Future Work

The most significant limitation is dataset size — 3,900 labelled KOIs is modest for deep learning, and the Genesis CNN's advantage over classical models would likely grow considerably with the larger candidate catalogues being produced by TESS and PLATO. The light curve pipeline was designed and implemented but not fully executed due to data availability constraints, and completing it remains the most immediate direction for future work — the Genesis architecture applied to raw Kepler flux time series is expected to achieve ROC-AUC above 0.97 based on Visser et al. (2022). Beyond that, three further directions stand out: applying SMOTE oversampling to improve CANDIDATE class performance, replacing GlobalAveragePooling with a self-attention mechanism to let the model focus on the most informative feature positions, and combining the Genesis CNN and Random Forest outputs in an ensemble to get the best of both approaches.

CHAPTER 7

CONCLUSION

7. Conclusion

This project set out to answer one question: can a simplified Genesis CNN accurately vet exoplanet candidates from Kepler KOI data, and how does it stack up against classical machine learning models on a three-class task? Having trained and evaluated five models on the same dataset and test split, the answer is nuanced but clear.

Random Forest was the strongest overall performer at 89.4% accuracy, F1 Macro 0.778, and ROC-AUC 0.955 — a result that sits comfortably within the 87–90% range reported by Armstrong et al. (2021) for similar tabular KOI classification tasks, and one that reflects the well-established strength of tree-based ensembles on small structured datasets. The Genesis CNN reached 85.7% accuracy and outperformed the Baseline CNN on every primary metric, with the improvement in CONFIRMED F1 from 0.842 to 0.890 providing the clearest evidence that the dual-branch design is doing something genuinely useful — the global branch is capturing cross-group feature relationships that a single-branch kernel-3 architecture simply cannot access. The CANDIDATE class was the hardest to classify across all five models, and honestly that was expected — these are signals that even experienced astronomers cannot agree on, and no amount of model complexity will fully resolve that ambiguity without better data.

The gap between the Genesis CNN and Random Forest is not an architectural problem. Augmentation experiments showed that doubling the training data pushed Genesis CNN accuracy up to 87–88%, narrowing the gap considerably and confirming that the model is constrained by data volume rather than design. As larger candidate catalogues from TESS and PLATO become available, this architecture should become increasingly competitive. The most immediate next step is completing the light curve pipeline — applying the same Genesis design to raw Kepler flux time series for binary classification — which is expected to achieve ROC-AUC above 0.97 based on published results from Visser et al. (2022). Beyond that, SMOTE oversampling for the CANDIDATE class and a self-attention mechanism in place of GlobalAveragePooling are the most promising directions for improving on what has been achieved here.

References

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., ... Zheng, X. (2015). *TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems*. <https://www.tensorflow.org>
- Akeson, R. L., Chen, X., Ciardi, D., Crane, M., Good, J., Harbut, M., Jackson, E., Kane, S. R., Laity, A. C., Leifer, S., Lynn, M., McElroy, D. L., Merchant, M., Ramírez, S., Ratzloff, J., ... Zhang, A. (2013). The NASA Exoplanet Archive: Data and Tools for Exoplanet Research. *Publications of the Astronomical Society of the Pacific*, 125(930), 989–999. <https://doi.org/10.1086/672273>
- Ansdell, M., Ioannou, Y., Osborn, H. P., Sasdelli, M., Smith, J. C., Caldwell, D., Jenkins, J. M., Räisänen, C., Gantert, G., Ehrig, H., & Vanderburg, A. (2018). Scientific Domain Knowledge Improves Exoplanet Transit Classification with Deep Learning. *The Astrophysical Journal Letters*, 869(1), L7. <https://doi.org/10.3847/2041-8213/aaf23b>
- Armstrong, D. J., Gamper, J., & Damoulas, T. (2021). Exoplanet validation with machine learning: 50 new validated Kepler planets. *Monthly Notices of the Royal Astronomical Society*, 504(4), 5327–5344. <https://doi.org/10.1093/mnras/staa2498>
- Borucki, W. J., Koch, D., Basri, G., Batalha, N., Brown, T., Caldwell, D., Caldwell, J., Christensen-Dalsgaard, J., Cochran, W. D., DeVore, E., Dunham, E. W., Dupree, A. K., Gautier, T. N., Geary, J. C., Gilliland, R., Gould, A., Howell, S. B., Jenkins, J. M., Kondo, Y., ... Welsh, W. J. (2010). Kepler Planet-Detection Mission: Introduction and First Results. *Science*, 327(5968), 977–980. <https://doi.org/10.1126/science.1185402>
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, 321–357. <https://doi.org/10.1613/jair.953>
- Cortes, C., & Vapnik, V. (1995). Support-Vector Networks. *Machine Learning*, 20(3), 273–297. <https://doi.org/10.1007/BF00994018>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. <https://jmlr.org/papers/v12/pedregosa11a.html>
- Shallue, C. J., & Vanderburg, A. (2018). Identifying Exoplanets with Deep Learning: A Five-Planet Resonant Chain around Kepler-80 and an Eighth Planet around Kepler-90. *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>

Thompson, S. E., Coughlin, J. L., Hoffman, K., Mullally, F., Christiansen, J. L., Burke, C. J., Bryson, S., Batalha, N., Haas, M. R., Catanzarite, J., Rowe, J. F., Barentsen, G., Caldwell, D. A., Clarke, B. D., Jenkins, J. M., Li, J., Latham, D. W., Lissauer, J. J., Mullally, S. E., ... Twicken, J. D. (2018). Planetary Candidates Observed by Kepler. VIII. A Fully Automated Catalog Based on Data Release 25. *The Astrophysical Journal Supplement Series*, 235(2), 38. <https://doi.org/10.3847/1538-4365/aab4f9>

Visser, K., Bosma, B., & Postma, E. (2022). Exoplanet detection with Genesis. *Journal of Astronomical Instrumentation*, 11(3), 2250011. <https://doi.org/10.1142/S2251171722500118>